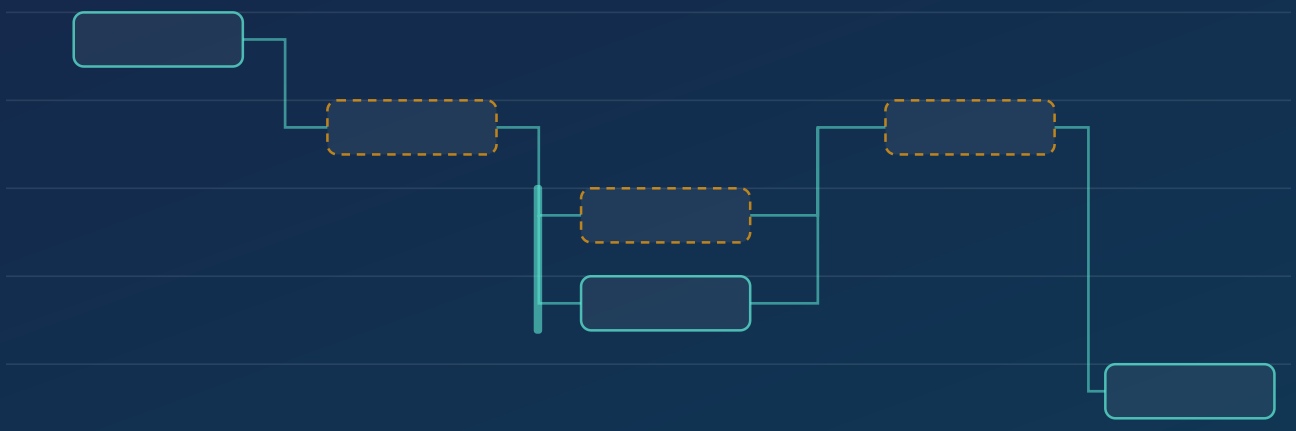
PRODUCT REQUIREMENTS DOCUMENT

BPMap for Workday

See, simulate and govern Workday business processes — privately, in the browser.

VERSION	1.0 — Draft for review
DATE	10 November 2026
OWNER	Rishabh (Product & Engineering)
STATUS	Draft. Open questions are listed in §19
SCOPE	MVP → Parity → Differentiation → Team & Enterprise (24 weeks)



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Document control

Product	BPMap for Workday *(working name; evolves the current "Workday Flowchart AI" prototype)*
Document	Product Requirements Document (PRD)
Version	1.0 — Draft for review
Date	10 November 2026
Owner	Rishabh (Product & Engineering)
Status	Draft. Open questions are listed in §19
Source inputs	Codebase review of `workday-flowchart-ai`; LinkedIn post by Rick Leunisse (Nov 2026); hands-on teardown of incubane.com/tools/bp-visualizer; `SCALE_UP_PLAN.md`

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Contents

1 Executive summary
2 Problem statement
3 Market & competitive analysis
4 Current-state assessment (prototype v0)
5 Goals, non-goals & success metrics
6 Users & personas
7 User journeys & key workflows
8 Functional requirements
9 Diagram notation specification
10 Non-functional requirements
11 Solution architecture
12 AI design principles & guardrails
13 Health-check rule catalogue
14 Release plan & roadmap
15 Pricing & packaging (proposed)
16 Go-to-market
17 Analytics & instrumentation
18 Risks & mitigations
19 Open questions & assumptions
Appendix A — Prototype issue register
Appendix B — Glossary

1. Executive summary

Every Workday customer and implementation partner has the same problem: **business process (BP) definitions are hard to read.** A Change Job or Hire definition has 15–40 steps, most with entry conditions, parallel stages, multiple security groups and subprocesses. Teams redraw them in Visio for workshops, audits, training and release testing. This takes hours, the drawings go stale, and they are often wrong.

The current prototype (*Workday Flowchart AI*) turns a screenshot of the Steps table into a swimlane chart using Gemini. It proves the need, but it has three problems that stop it scaling:

1. **Privacy.** Tenant configuration is sent to a public LLM, and the API key is exposed in the browser.
2. **Correctness.** Conditions are drawn as "No → return to initiation" (Workday *skips* the step), parallel steps are drawn as serial, and most step types are lost.
3. **Depth.** It produces a viewport PNG and nothing else: no views, scenarios, health check, compare or editable exports.

Meanwhile a free, **local-only, non-AI** competitor (Incubane BP Visualizer, promoted on LinkedIn in Nov 2026) has set the baseline: Excel import, correct Workday semantics, three audience views, scenarios, a light health check, customisation overlays, and PDF/PPTX/draw.io export.

Our strategy: deterministic core, optional AI, and value beyond the diagram.

- **Match the baseline, but privately and accurately.** Parse the Workday Excel export in the browser, model Workday semantics properly, and render the readable notation that is becoming the norm.
- **Go further than a diagram.** Generate test plans from scenarios, run a real health/controls rule engine (including segregation of duties), compare versions and tenants, link subprocesses, and export BPMN 2.0.
- **Use AI only where it helps, and only with consent.** Opt-in and redacted, working on the structured model rather than screenshots. Uses: plain-English condition wording, SOPs, change summaries for CAB, and Q&A.
- **Grow from a free tool into a team product.** Shared library, versioning, comments, SSO and self-hosting for partners and enterprise HRIS/Finance teams.

Headline targets (12 months after Phase 2): 2,000 monthly active users, 30% import-to-export activation, 100% parse fidelity on a 50-definition test corpus, and 40 paying teams.

2. Problem statement

2.1 The problem

Who	Pain today	Cost
HRIS / Finance systems analysts	Have to read raw BP definitions in Workday's grid UI or export them to Excel; no visual	Hours per process; misread conditions cause defects
Workday consultants (partners)	Redraw every BP in Visio for design workshops and handover docs	2–6 hours per BP; inconsistent quality; rework after each change
Process owners & executives	Can't answer "who approves what, and when?"	Slow decisions; approvals nobody questions
Internal audit / SOX	Need evidence of approval controls and segregation of duties	Manual walkthroughs; audit findings

Who	Pain today	Cost
Test & release managers	Workday ships two feature releases a year; BP changes need regression	Missed condition paths; production incidents

2.2 Why now

- **Demand is visible.** The Incubane LinkedIn launch got about 70 reactions and comments like "I was literally looking for something like that today."
- **The baseline is now free.** A tool that only draws a diagram won't win. Value has to come from simulation, testing, controls, comparison and collaboration.
- **Buyers now ask about AI governance.** Teams want AI help but will not upload tenant configuration casually. Opt-in, redacted, structured-data AI is a real differentiator.

2.3 Problem statement

> *HRIS teams and Workday consultants need a fast, trustworthy way to turn Workday BP definitions into readable, correct, audience-appropriate diagrams and downstream artefacts (test plans, controls evidence, change summaries) without exposing tenant configuration.*

3. Market & competitive analysis

3.1 Competitive landscape

Alternative	Strengths	Weaknesses
Manual Visio / draw.io / Lucid	Flexible, familiar	Slow, stale, error-prone, no semantics
Workday native BP view	Always current, authoritative	A grid, not a flow; hard to read; no audience views or exports
Incubane BP Visualizer (free)	Local-only; correct semantics; 3 views; scenarios; Health tips; overlays; PDF/PPTX/draw.io	Account needed to download; condition logic and security policy out of scope; subprocesses are separate tabs; 4 heuristic tips; no BPMN; no collaboration; no compare found on the web version
Process-mining / BPM suites (Signavio, Celonis, ARIS)	Enterprise governance, BPMN	Expensive; no native Workday definition import; heavy
Our prototype (v0)	Accepts screenshots; editable canvas	Cloud LLM; wrong semantics; PNG only

3.2 Positioning

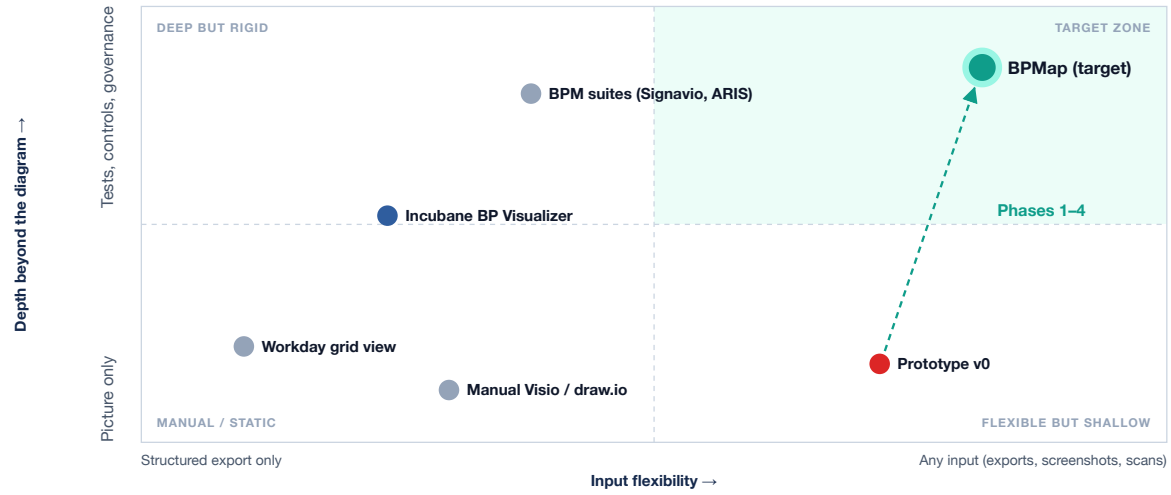


Figure 1 — Competitive positioning (illustrative)

3.3 Incubane BP Visualizer teardown (tested 10 Nov 2026)

Tested with the built-in *Change Job (Default Definition)* sample: 18 steps, 12 stages, 16 conditional, 9 security groups.

Area	Observation	Implication for us
Input	Excel from *View Definition → Export*; uses the Steps and Notifications sheets; several files, one tab each; in-app "how to export" guide	Table stakes (IMP-01..05, IMP-11)
Privacy	"Nothing is uploaded"; only site, cookieless analytics and fonts on the network; state in `localStorage`	Table stakes; we should *prove* it (NFR-PRIV-01)

Area	Observation	Implication for us
Notation	Columns = order letters; lanes = security groups; system lanes at bottom; **dashed border + C-badge for conditional steps**; condition list below; diamonds only on approvals; "No, skip"; exceptions explained once in the legend	Adopt and refine (VIS-01..06)
Views	Executive (stage cards), Process owner (swimlanes), Configurator (+ due, alternate, routing); Focus on a role	Adopt (VW-01..04, VIS-08)
Scenario	Yes / No / Not set per condition; skipped steps disappear; empty lanes collapse	Adopt, then extend with presets and test plans (SCN-*)
Health	4 informational heuristics	Replace with a rule engine with severities (HLT-*)
Customise	Lane names, condition wording, notes, hide step, downstream systems; **overlay survives re-import**	Adopt; key overlays to stable IDs (CUS-*)
Export	PDF, editable PPTX, draw.io, PNG, SVG; **free account required**	Match the formats; keep basic formats free without an account (EXP-*)
Gaps	No condition logic, no security policy, no linked subprocesses, no compare found, no BPMN, no collaboration	Our differentiators (§8)

4. Current-state assessment (prototype v0)

The prototype is a ~1,300-line React 19 + Vite + TypeScript app created in Google AI Studio. It uses React Flow for the canvas and Gemini 2.5 Flash for extraction.

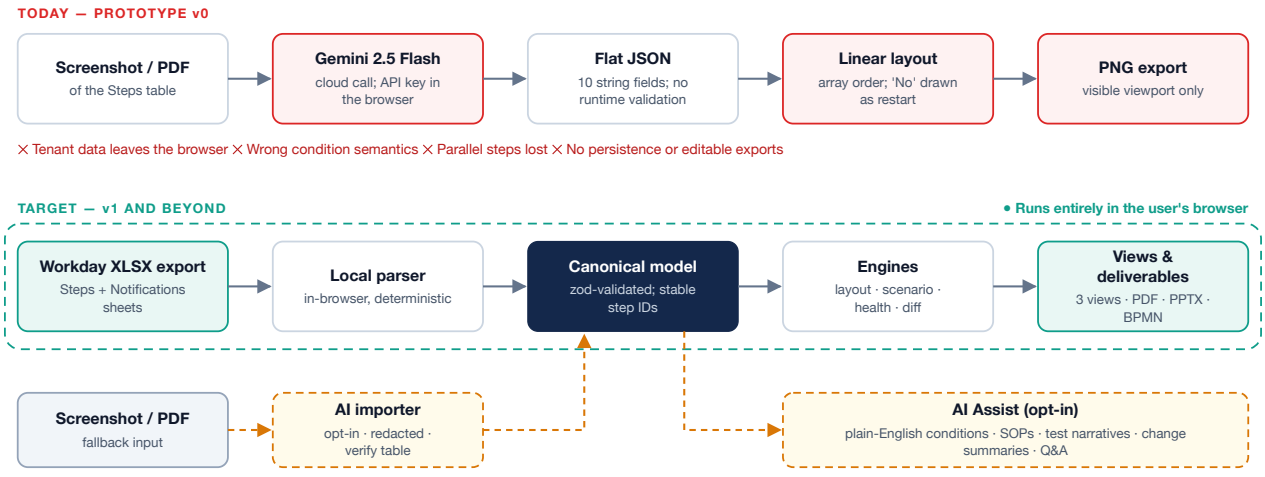


Figure 2 — Current pipeline vs target pipeline

What to keep: the extract → verify → regenerate loop, structured LLM output, React Flow, Tailwind, and screenshot input (as an opt-in fallback).

What must change: see the full issue register in Appendix A. The most severe items:

ID	Issue	Severity
D1	A false condition is drawn as "revert to initiation"; Workday skips the step	Critical
E1	The Gemini API key is read in client code; it can't be kept secret	Critical
E2	Tenant data is sent to a third-party LLM without consent	Critical
D2	Order letters ignored; parallel steps drawn as serial	High
D3/D4	Only 4 step types; approval nodes don't say who approves	High
E4	PNG export captures only the visible viewport	High
E5	Canvas and table are separate sources of truth; edits are lost on regenerate	High

5. Goals, non-goals & success metrics

5.1 Goals

#	Goal	Measure
G1	Produce a **correct** diagram from any Workday BP export in under a minute	Median time from import to first render < 60 s; 100% fidelity on the fixture corpus
G2	**Keep tenant data private by default**	0 bytes of file content leave the browser in default mode (automated test)
G3	Serve **three audiences** from one model	≥ 50% of exports use a non-default view
G4	Go **beyond the diagram** (tests, controls, compare)	≥ 25% of Pro users generate a test plan or controls matrix monthly
G5	Build a **sustainable business**	40 paying teams within 12 months of Phase 2

5.2 Non-goals (v1)

- Editing or deploying BP definitions back into Workday (read-only only).
- Process mining from transaction logs.
- Building a general-purpose diagram editor.
- Mobile editing (tablet is read-only).
- Supporting non-Workday HCM/ERP process formats.

5.3 Success metrics

Type	Metric	Target
North star	**Definitions delivered**: weekly exports + shares + test plans	1,500 / week by month 12
Activation	% of new users who import a file and export within the first session	≥ 30%
Quality	Parse fidelity on the golden corpus	100% (release gate)
Quality	User-reported diagram errors per 100 imports	< 1
Engagement	Weekly active users (WAU) / monthly active users (MAU)	≥ 35%
Monetisation	Free → Pro conversion	3–5%
Trust	Security reviews passed without blocking findings	100%
Satisfaction	NPS from design partners	≥ 40

6. Users & personas

Persona	Role	Primary jobs	Key features	Tier
Priya — HRIS Analyst	Customer systems team	Document as-is BPs; impact analysis before changes; release regression	Import, Configurator view, Compare, Health, Test plan	Free → Team
Marco — Workday Consultant	Partner / independent	Design workshops, deliverables, handover packs	Packs, branded PPTX/PDF, Customise, Scenarios	Pro
Dana — Process Owner	HR / Finance business lead	Understand and approve the process design	Process-owner view, plain-English conditions, SOP	Free (viewer)
Sam — Executive sponsor	VP HR / CFO office	One-slide overview; cycle-time hotspots	Executive view	Free (viewer)
Alex — Internal Auditor	SOX / controls	Evidence of approvals and segregation of duties	Controls matrix, SoD rules, security policy import	Team / Enterprise
Jo — Test Lead	Release / QA	Test every condition path after each Workday release	Scenario presets, test-plan generator, Compare	Pro / Team

7. User journeys & key workflows

7.1 End-to-end workflow

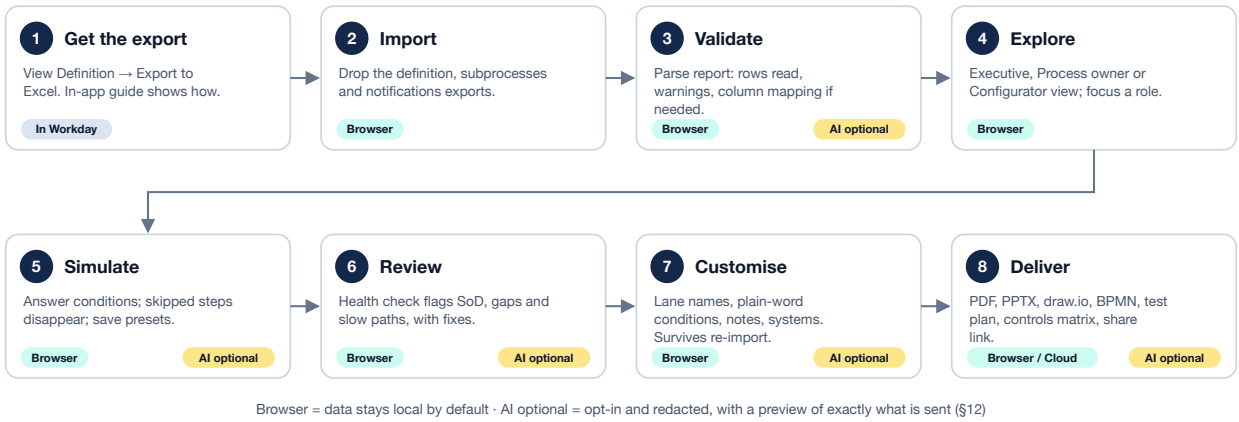


Figure 3 — End-to-end user workflow (AI Assist optional at steps 3, 5, 6, 7 and 8)

7.2 Key user stories

ID	As a…	I want to…	So that…	Priority
US-01	HRIS analyst	drop the Change Job export and see a correct swimlane in under a minute	I can explain the process in a meeting today	P0
US-02	Consultant	switch between Executive, Process owner and Configurator views	each audience sees the right level of detail	P1
US-03	Process owner	answer "promotion? works-council country?" and see only the steps that run	I understand the real path for a case	P1
US-04	Test lead	generate test cases for every meaningful condition combination	regression after a Workday release covers every path	P2
US-05	Auditor	see where an initiator could approve their own transaction	I can evidence segregation of duties	P2
US-06	Analyst	compare the Sandbox and Production definitions	I know exactly what will change at go-live	P2
US-07	Consultant	rewrite conditions in plain words and keep them after re-import	my customisations aren't lost each iteration	P1
US-08	Consultant	export an editable PowerPoint with my client's branding	I can drop it straight into the deliverable	P1 / P2
US-09	Analyst	open the Propose Compensation subprocess from the step that calls it	I can follow the whole end-to-end process	P2
US-10	Executive	get a one-slide view of stages and owners	I understand the process in a minute	P1
US-11	Security reviewer	verify the tool sends no tenant data by default	I can approve it for use	P0
US-12	Analyst with only a screenshot	still produce a draft diagram, with consent	I'm not blocked when I can't export	P2

8. Functional requirements

Priority key: **P0** = MVP (Phase 1) · **P1** = parity (Phase 2) · **P2** = differentiation (Phase 3) · **P3** = team/enterprise (Phase 4).

8.1 Import (IMP)

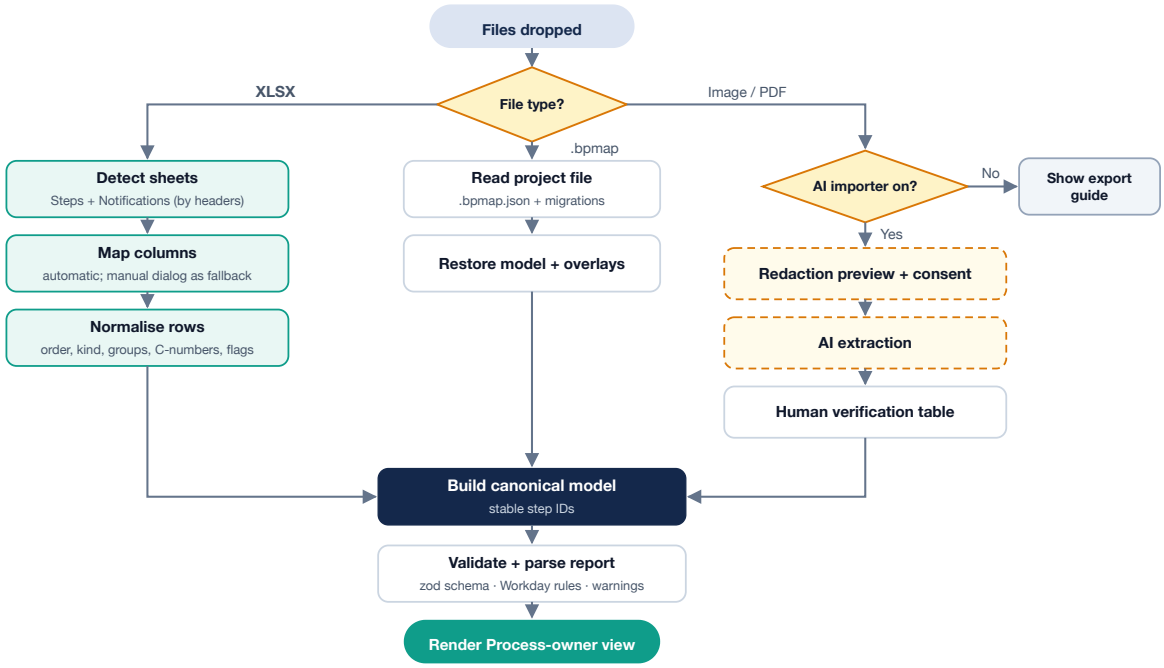


Figure 4 — Import and parsing pipeline

ID	Requirement	Priority	Acceptance criteria
IMP-01	Import the Workday *View Definition* Excel export (.xlsx) by drag-and-drop or file picker, parsed entirely in the browser (in a Web Worker)	P0	Golden files for Change Job, Hire, Terminate, Create Journal and Create Supplier Invoice parse with 100% step fidelity; an automated test confirms no network request carries file content
IMP-02	Detect the Steps and Notifications sheets by header signature (not sheet name or position); ignore other sheets with an info note	P0	Works on all fixtures, including renamed and reordered sheets
IMP-03	Map columns automatically, with a manual mapping dialog as fallback and saved mapping profiles	P0	An unknown layout opens the dialog; a saved profile is reused silently on the next import
IMP-04	Multi-file import: a definition plus subprocess definitions plus a notifications export, grouped into one project	P1	Up to 20 files; subprocesses auto-linked to the calling step by task name; unmatched files listed
IMP-05	Show a parse report after every import: rows read, steps built, conditions, unknown types, warnings with row references	P0	Report is reachable from the toolbar; warnings link to rows
IMP-06	Re-import a newer export of the same definition; keep overlays and flag changes	P1	Overlays on steps with an unchanged stable key are preserved 100%; orphaned overlays listed for review

ID	Requirement	Priority	Acceptance criteria
IMP-07	Save and load the project file (`.bpmap.json`) with schema version and migrations	P0	Files from older schema versions load after migration; round-trip is lossless
IMP-08	Opt-in AI vision importer for screenshots and PDFs; output always goes through the verification table	P2	Off by default; redaction preview shown before sending; verification required before render
IMP-09	Import the condition-rule report to attach the real rule logic to each C-number	P2	Logic shown in the Step and Scenario panels; *report format to be validated with design partners*
IMP-10	Import the BP security policy: who may initiate, approve, cancel, rescind or correct	P2	Allowed groups shown on the Step panel and used by SoD rules
IMP-11	In-app "How to get the export" guide (3 steps, with screenshots)	P0	Linked from the empty state and from import errors

8.2 Process model & Workday semantics (MOD)

ID	Requirement	Priority	Acceptance criteria
MOD-01	Canonical model (§11.3), validated with zod and versioned	P0	All importers produce it; invalid data never reaches the renderer
MOD-02	Stable step IDs: `{order}-{kind}-{slug(task)}` plus a collision suffix; never the array index	P0	The same export always produces the same IDs; IDs survive row re-ordering
MOD-03	**Order semantics:** letters run in sequence; steps with the same letter run in parallel; suffixed orders (e.g. `f1`) are their own stage between `f` and `g`	P0	Fixture with 6 parallel steps renders as one stage
MOD-04	**Condition semantics:** a step runs only if all its entry conditions are true; false means **skip**, never "return to initiation"	P0	No "revert" edges anywhere; "No, skip" label on decisions
MOD-05	Step-kind catalogue: initiation, approval, approval chain, consolidated approval chain, action, to do, review, checklist, questionnaire, integration, service, sub-process; unknown types kept and flagged	P0	Each kind has a distinct shape or label; unknown types appear in the parse report
MOD-06	Steps with several groups (e.g. Payroll Interface Partner + Payroll Partner) and "all must act"	P0	Primary lane plus an "Also:" list; the "all must act" badge is visible
MOD-07	Configurator attributes: completion step, due date (relative / effective-date), alternate, routing, restriction, run-as	P1	Shown in Configurator view and the Step panel
MOD-08	Workday-owned vs customer-owned flag on each condition	P0	Shown in the condition list; used by health rule HC-07
MOD-09	Notifications attached to the process and to steps	P1	Listed in Configurator view and the Step panel

8.3 Visualisation & notation (VIS)

ID	Requirement	Priority	Acceptance criteria
VIS-01	Swimlane layout: **columns = order stages, lanes = security groups** (order of first appearance); Workday-service and integration lanes at the bottom	P0	Matches the notation in §9
VIS-02	Parallel split and join bars, with an "N in parallel" column header	P0	Present wherever a stage has more than one step

ID	Requirement	Priority	Acceptance criteria
VIS-03	Conditional steps drawn with a dashed outline and C-badge, plus a numbered condition list under the diagram; diamonds only on approvals or when toggled	P0	No diamond per condition by default
VIS-04	Exceptions (send back, deny, cancel, rescind, correct) explained once in the legend, not drawn as arrows	P0	Legend text included in every export
VIS-05	Automatic layout (ELK.js) with no overlapping nodes or edges through nodes, for definitions up to 80 steps	P0	Visual regression tests on the corpus
VIS-06	A legend that is always present and included in exports	P0	—
VIS-07	Zoom, fit, full screen, minimap, hide panel	P0	—
VIS-08	Focus on one role: dim every other lane	P1	—
VIS-09	Collapse empty lanes and stages when a scenario hides steps	P1	—
VIS-10	Accessible colours (WCAG 2.2 AA); meaning never carried by colour alone	P0	Automated contrast check; shape plus text for every state
VIS-11	Manual layout nudges, stored in the overlay	P1	Survive re-import for unchanged steps

8.4 Audience views (VW)

ID	Requirement	Priority	Acceptance criteria
VW-01	**Executive:** one card per stage, parallel steps grouped, "only if" tags; fits one 16:9 slide for up to 15 stages	P1	Rendered PPTX slide passes visual review
VW-02	**Process owner:** swimlane diagram with plain-language labels	P0	Default view
VW-03	**Configurator:** adds due date, alternate, routing, restriction, run-as, notifications and Workday task names	P1	All MOD-07/09 fields visible
VW-04	The chosen view is saved per definition and applied to exports	P1	—

8.5 Scenario simulation & test planning (SCN)

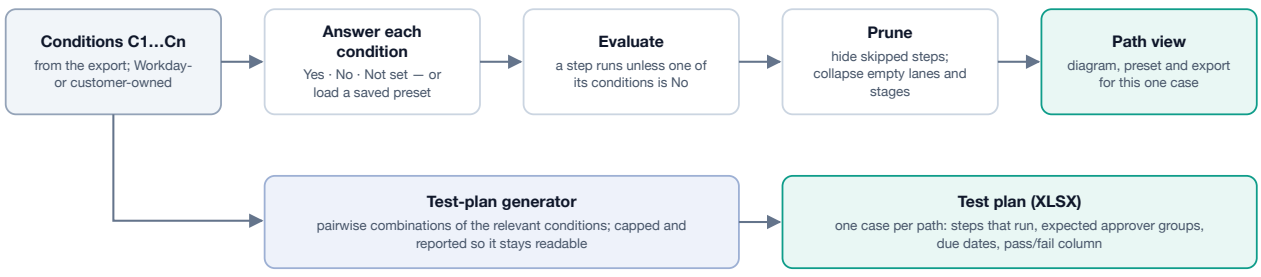


Figure 5 — Scenario simulation and test-plan generation

ID	Requirement	Priority	Acceptance criteria
SCN-01	Each condition can be set to Yes, No or Not set; skipped steps are hidden; the header shows the active scenario	P1	Removing a step re-routes edges with no gaps

ID	Requirement	Priority	Acceptance criteria
SCN-02	Toggle to draw a condition as a yes/no decision diamond	P1	—
SCN-03	Named scenario presets (e.g. "Promotion – Netherlands") saved per definition and exportable	P1	A preset can be applied to exports
SCN-04	Compare two scenarios side by side	P2	Differences highlighted
SCN-05	**Test-plan generator:** pairwise combinations of the relevant conditions produce test cases with steps, expected approver groups and due dates; exported as XLSX	P2	Every step is covered by at least one test case; combinatorial explosion is capped and reported

8.6 Health check & controls (HLT, CTL)

ID	Requirement	Priority	Acceptance criteria
HLT-01	Rules engine (catalogue in §13) runs on every model change; findings grouped as Critical, Warning or Info	P1 (HC-01..06), P2 (rest)	Under 200 ms for 80 steps
HLT-02	Each finding has an explanation, affected steps (click to highlight), a suggested fix, and "dismiss with reason" (stored in the overlay)	P1	—
HLT-03	Health score (0–100) per definition and per pack	P2	Weighting documented
HLT-04	Rule thresholds configurable per workspace	P3	—
CTL-01	Segregation-of-duties checks using the imported security policy	P2	Initiator-can-approve paths flagged as Critical
CTL-02	Controls matrix export (XLSX): control ID, BP, step, group, condition, evidence reference	P2	Opens cleanly in Excel
CTL-03	Map steps to client control IDs (stored in the overlay)	P2	—

8.7 Customisation (CUS)

ID	Requirement	Priority	Acceptance criteria
CUS-01	Overlay layer: lane renames, step labels, notes ("N" badge), hide step, plain-word condition text, downstream systems per step and on completion	P0 (labels, notes, hide) · P1 (rest)	The model itself is never changed by an overlay
CUS-02	The overlay survives re-import (keyed on stable IDs)	P1	See IMP-06
CUS-03	Save and load customisations as a separate file to share with colleagues	P1	—
CUS-04	Multi-level undo and redo (at least 50 steps)	P0	Typing in a field counts as one undo step
CUS-05	Branding templates: logo, colours and footer text for exports	P2	Applied to PDF and PPTX

8.8 Compare (CMP)

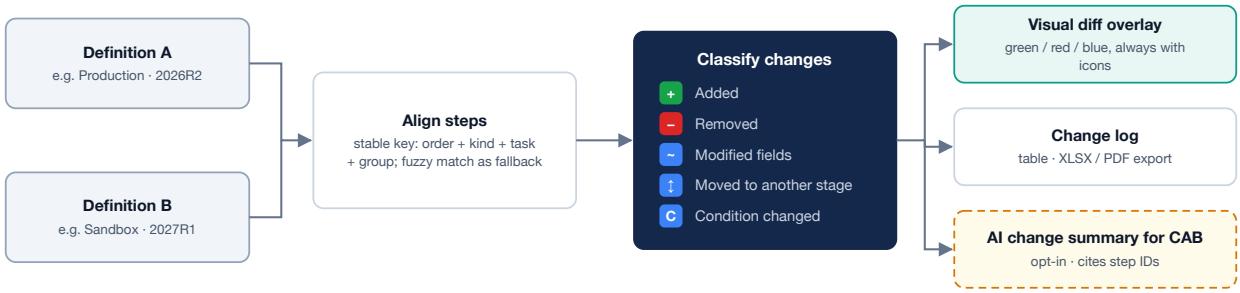


Figure 6 — Compare (version vs version, tenant vs tenant)

ID	Requirement	Priority	Acceptance criteria
CMP-01	Compare two definitions: two versions or two tenants	P2	Matching precision ≥ 98% on the diff fixtures
CMP-02	Visual diff overlay (added = green, removed = red, modified = blue; amber stays reserved for conditions, see Design System D-03) plus a change log table	P2	Colours paired with icons and text
CMP-03	Export the change log to XLSX or PDF	P2	—
CMP-04	AI change summary in business language (opt-in)	P2	Every statement cites step IDs

8.9 Packs & subprocesses (PCK)

ID	Requirement	Priority	Acceptance criteria
PCK-01	Group definitions into a pack (e.g. the Hire family)	P1	Tabs per definition
PCK-02	Drill down from a sub-process step into its definition and back (breadcrumb)	P2	Works for nesting up to 3 levels deep
PCK-03	Pack export: cover, contents, a section per definition, combined condition list and health summary	P1	—

8.10 Export (EXP)

ID	Requirement	Priority	Acceptance criteria
EXP-01	PNG and SVG of the **full** diagram (not the viewport), at 2×	P0	Bounds computed from nodes
EXP-02	Vector PDF: multi-page, A4/Letter/A3, legend and condition list included	P1	Text selectable
EXP-03	PPTX with **native editable shapes**: one slide per view plus a conditions slide	P1	Every box and connector editable in PowerPoint
EXP-04	draw.io XML (opens in diagrams.net; save-as-Visio path documented)	P1	Lanes and containers preserved
EXP-05	Mermaid	P2	—

ID	Requirement	Priority	Acceptance criteria
EXP-06	BPMN 2.0 XML: lanes, parallel gateways, conditional flows	P2	Opens in Camunda Modeler and Signavio
EXP-07	Copy image to clipboard	P0	—
EXP-08	XLSX: steps, conditions, health findings, test plan	P1 / P2	—
EXP-09	Free tier exports PNG, SVG and PDF **without an account**; account needed for Pro formats	P1	Pricing decision in §15

8.11 AI Assist (AI)

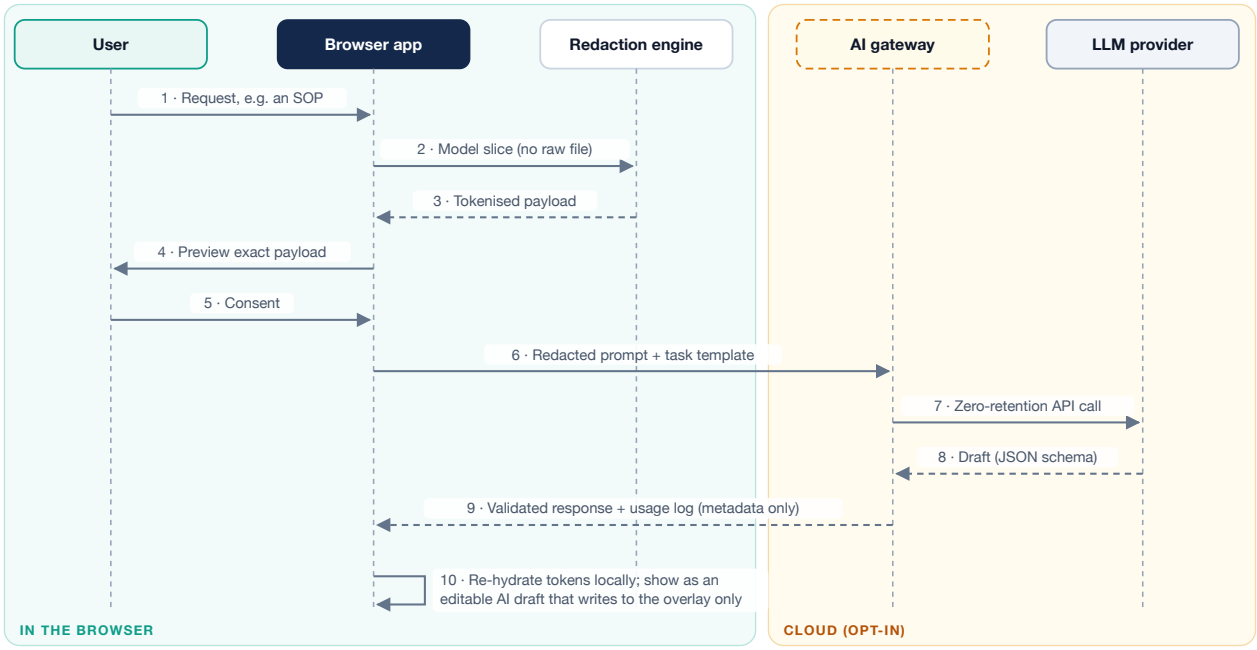


Figure 7 — AI Assist data flow with redaction and consent

ID	Requirement	Priority	Acceptance criteria
AI-01	Off by default; workspace-admin and user toggles; a persistent indicator while on	P2	Default install makes no AI calls
AI-02	Redaction and preview: security group names, people, IDs and custom condition text can be tokenised; the preview shows the exact payload	P2	The payload in the preview is byte-identical to what is sent
AI-03	Draft plain-English wording for conditions (uses rule logic if IMP-09 is loaded)	P2	Accepted text goes into the overlay, never the model
AI-04	SOP / narrative for each audience view	P2	Exports to DOCX / PDF
AI-05	Test-scenario narratives and expected results (adds to SCN-05)	P2	—
AI-06	Change summary for CAB from a diff (CMP-04)	P2	Cites step IDs
AI-07	"Ask the process": Q&A grounded only on the loaded model, with step citations	P2	Refuses when the answer isn't in the model
AI-08	Every AI output is labelled, editable, and never changes the model silently	P2	—

ID	Requirement	Priority	Acceptance criteria
AI-09	Provider-agnostic gateway; zero-retention providers only; logs metadata only (no content)	P2	Documented data-processing terms
AI-10	Bring-your-own key (Pro); self-hosted model option (Enterprise)	P3	—

8.12 Collaboration & enterprise (COL, ENT)

ID	Requirement	Priority	Acceptance criteria
COL-01	Workspace library with version history per definition	P3	Any version can be restored
COL-02	Read-only share links with expiry and revocation	P3	Viewers need no account
COL-03	Comments and @mentions on steps	P3	Email notifications
ENT-01	SSO (SAML 2.0 / OIDC) and SCIM provisioning	P3	Okta and Entra ID tested
ENT-02	Roles: owner, editor, viewer	P3	—
ENT-03	Audit log of imports, exports, shares and AI calls	P3	Exportable
ENT-04	Data residency (EU / US) and a self-hosted container build	P3	—
ENT-05	Tenant connector: scheduled pull of BP definitions via a custom report (RaaS)	P3	**Validate feasibility first**

9. Diagram notation specification

The notation follows one principle: **show everything that changes the path, and explain everything else once.**

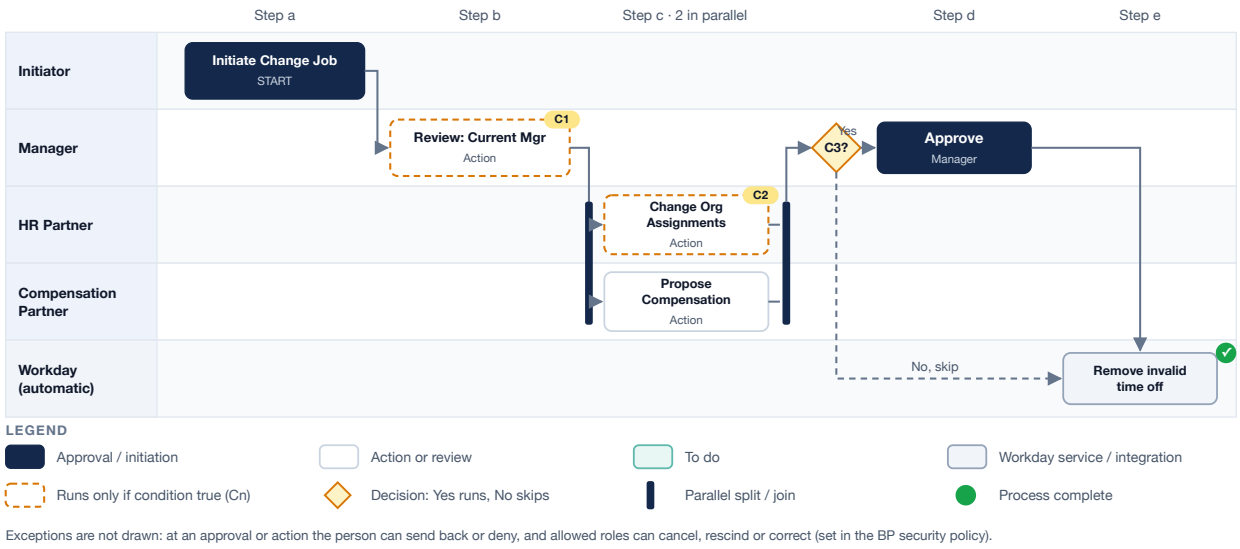


Figure 8 — Notation reference: stages as columns, security groups as lanes

Element	Rendering	Rule
Stage	Column headed "Step a", "Step b"… and "· N in parallel"	One column per distinct order value
Lane	Horizontal band labelled with the security group	Ordered by first appearance; system lanes last
Initiation	Navy box with a "START" tag	Always the first stage
Approval / approval chain	Navy filled box	Label = task or "Approve", plus the group
Action / review	White box, solid outline	Label = Workday task
To do	Light-teal box	Label = to-do text
Workday service / integration	Grey box in the system lane	Label = service name
Conditional step	**Dashed amber outline + C-badge**	Condition text in the list under the diagram
Decision	Diamond "Cn?" with "Yes" and "No, skip"	Only on approvals by default, or when toggled
Parallel	Split and join bars	Wherever a stage has more than one step
Completion	Green check after the completion step	From the "complete" flag
Note	"N" badge	From the overlay
Exceptions	Legend text only	Send back, deny, cancel, rescind, correct

10. Non-functional requirements

ID	Category	Requirement
NFR-PRIV-01	Privacy	In default mode no file content, model or derived data leaves the browser. Enforced by a Content-Security-Policy `connect-src` allowlist and an automated network-capture test in CI
NFR-PRIV-02	Privacy	Analytics are cookieless and never contain content or file names
NFR-SEC-01	Security	No secrets in the client bundle; CI scans `dist/` for key patterns
NFR-SEC-02	Security	Strict CSP and Subresource Integrity (SRI); dependency scanning; XLSX parsed in a Web Worker with a 25 MB limit and zip-bomb guards
NFR-SEC-03	Security	AI gateway: authentication, rate limiting, payload size limits, prompt-injection guards (model content is treated as data), zero-retention providers
NFR-PERF-01	Performance	Parse and render a 60-step definition in < 2 s (P95) on a 2020-era laptop; 200 steps in < 5 s
NFR-PERF-02	Performance	Single-definition PDF export in < 5 s; 10-definition pack in < 30 s
NFR-A11Y	Accessibility	WCAG 2.2 AA; keyboard navigation between steps; a screen-reader step list as an alternative to the canvas
NFR-COMP	Compatibility	Latest two versions of Chrome, Edge, Safari and Firefox; desktop first; tablet read-only
NFR-REL	Reliability	Autosave to IndexedDB on every change; recovery after a crash; nothing lost on refresh
NFR-I18N	Internationalisation	English UI at launch; parser tolerant of localised headers (EN, DE, FR, NL, ES) by Phase 3
NFR-QUAL	Quality	Parser line coverage ≥ 90%; golden corpus ≥ 30 definitions by Phase 2 and ≥ 50 by Phase 3; visual regression tests on every PR
NFR-LEGAL	Legal	Nominative trademark use only, with the disclaimer on every page and export; no Workday logo

11. Solution architecture

11.1 Logical architecture

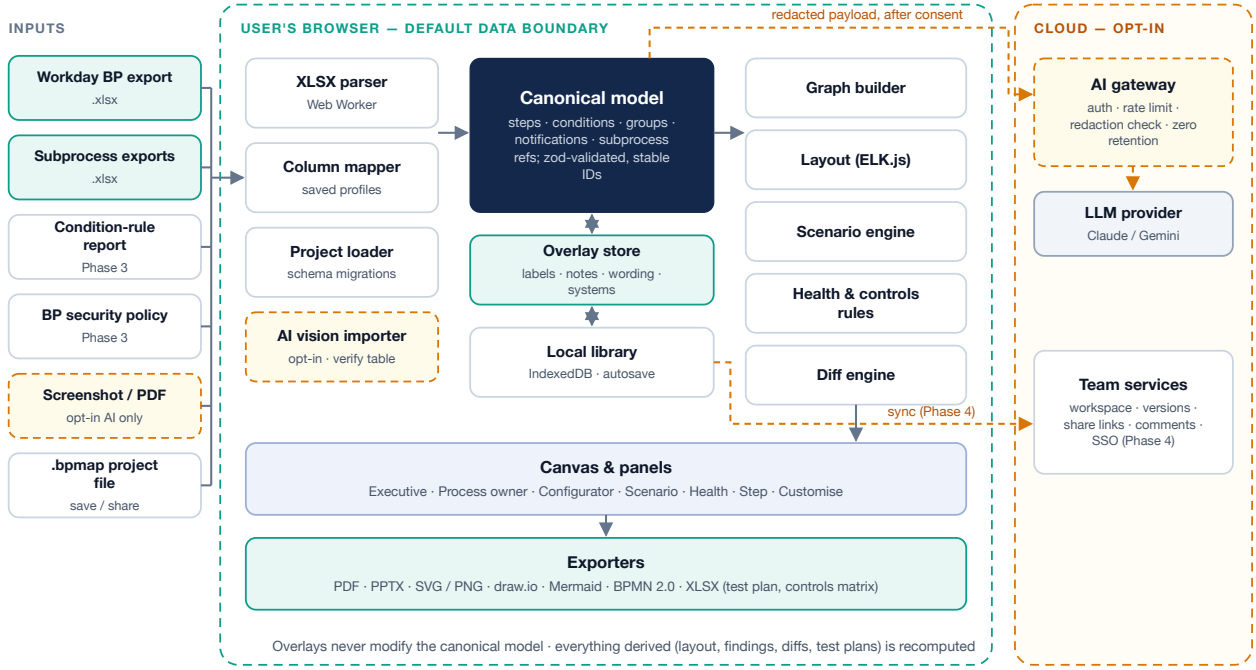


Figure 9 — Logical architecture and trust boundaries

11.2 Technology choices

Layer	Choice	Rationale
App shell	React 19 + Vite + TypeScript (keep)	Already in place; fast; the team knows it
Canvas	React Flow `@xyflow/react` v12 (keep)	Mature; custom nodes; good performance
Layout	**ELK.js** (layered, with partitions for lanes)	Handles parallel branches, lanes and edge routing
State	**Zustand + zundo**	Simple store with multi-level undo
Validation	**zod**	Runtime safety for every importer and the AI output
XLSX	**exceljs** (or SheetJS from its official distribution) in a Web Worker	Reads in the browser; worker keeps the UI responsive
Storage	**Dexie (IndexedDB)**	Local library, autosave
Export	`pptxgenjs`, `jspdf` + `svg2pdf.js`, hand-built draw.io / BPMN XML	Native editable outputs
Tests	Vitest (unit, golden files), Playwright (E2E, network capture, visual regression)	Supports the quality gates

Layer	Choice	Rationale
Backend (Phase 3+)	Small TypeScript service (e.g. Cloudflare Workers or Node) for the AI gateway; Postgres for team data (Phase 4)	Minimal surface area; can be self-hosted
AI	Provider-agnostic adapter; Claude or Gemini via zero-retention APIs; structured JSON outputs	Avoids lock-in; enterprise-acceptable

11.3 Data model

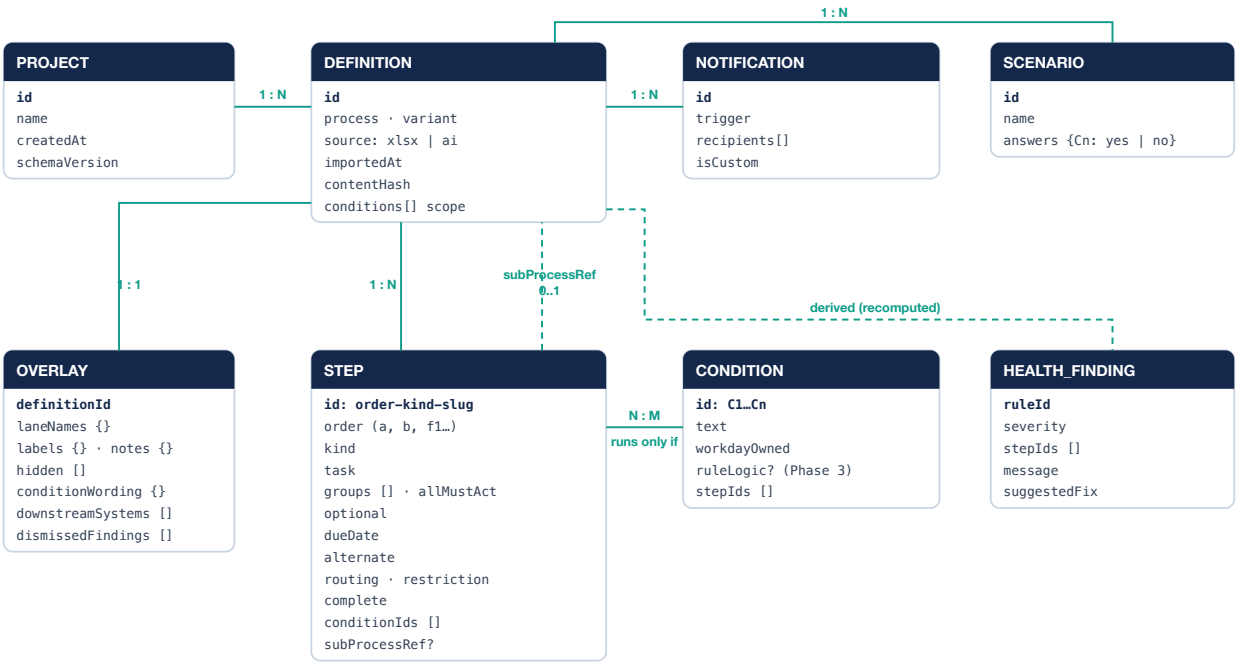


Figure 10 — Canonical data model

Design rules

1. **There is one source of truth: the canonical model.** The table, canvas and exports are all views of it.
2. **Overlays never change the model.** They are a separate layer keyed by stable IDs, so they survive re-import.
3. **Everything derived** (layout, health findings, diffs, test plans) is recomputed from the model and overlay, never stored as the source.

12. AI design principles & guardrails

Principle	Implementation
Opt-in, never default	Workspace policy and user toggle; persistent indicator while on
Structured in, structured out	Prompts are built from model slices; outputs validated with JSON schemas and zod
Minimum necessary data	Only the fields the task needs; redaction tokens for names, groups and IDs
Show before send	The preview is the exact payload; consent is logged

Principle	Implementation
Human in the loop	AI output is a labelled draft; accepting it writes to the overlay, never the model
Grounded and cited	Q&A and summaries cite step and condition IDs; refuse when the answer isn't in the model
No training, no retention	Zero-retention providers only; the gateway logs metadata only
Evaluated	Offline eval set per AI feature (wording, SOP, change summary); regression before every model upgrade

13. Health-check rule catalogue

ID	Rule	Severity	Data needed	Phase
HC-01	Approval step not restricted from the initiator (self-approval risk)	Critical	Restriction field (or security policy)	2
HC-02	Required approval or action with no alternate	Warning	Alternate	2
HC-03	Approval or action with no due date	Info	Due date	2
HC-04	Same group acts in consecutive serial stages (could run in parallel)	Info	Order, groups	2
HC-05	Group receives three or more tasks in one stage (consider a checklist or consolidated step)	Info	Order, groups	2
HC-06	Serial approval chain longer than N (default 4)	Warning	Order, kinds	2
HC-07	Customer-owned conditions to re-validate after each Workday release	Info	workdayOwned	2
HC-08	Approval placed after the completion step	Warning	Complete flag	3
HC-09	Called subprocess not loaded	Info	Sub-process refs	3
HC-10	Unknown or unsupported step type	Warning	Kind	3
HC-11	Duplicate step (same task and group in the same stage)	Warning	Task, group, order	3
HC-12	Optional approval on a Financials BP (control may be skipped)	Warning	Optional, BP family	3
HC-13	Initiating group can also cancel, rescind or correct (SoD)	Critical	Security policy (IMP-10)	3
HC-14	No completion notification to the initiator	Info	Notifications	3
HC-15	Condition referenced on many steps with diverging plain-word overrides	Info	Overlay	3

14. Release plan & roadmap

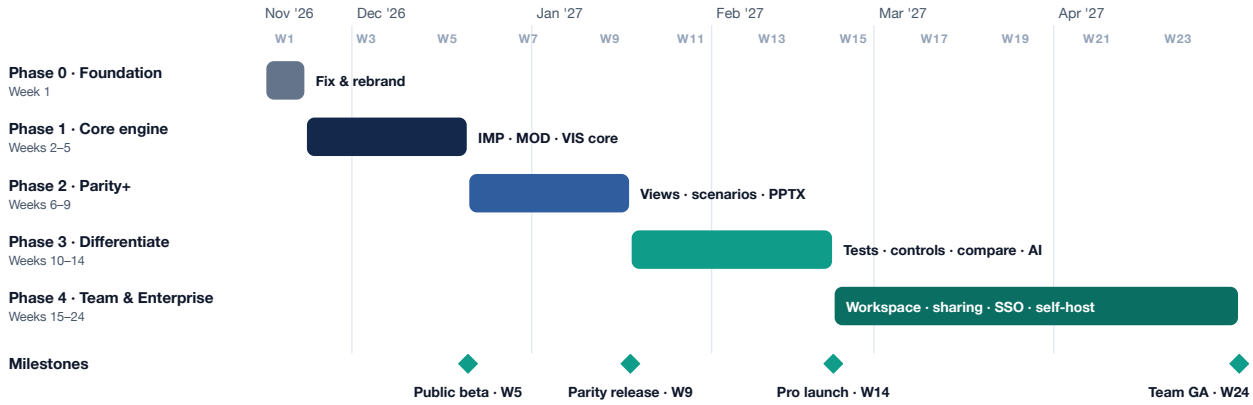


Figure 11 — Roadmap and milestones

Phase	Weeks	Scope (requirement IDs)	Exit criteria
0. Foundation	1	Install toolchain, git, CI; fix E1/E3/E4/E6/E10/E11; remove the client-side key; rebrand	`lint`, `test` and `build` pass; secret scan clean; no Workday logo
1. Core engine → *Public beta*	2–5	IMP-01/02/03/05/07/11 · MOD-01..06/08 · VIS-01..07/10 · VW-02 · CUS-01 (part)/04 · EXP-01/07	20+ anonymised definitions at 100% fidelity; network test green; 10 beta users
2. Parity+ → *Parity release*	6–9	IMP-04/06 · MOD-07/09 · VIS-08/09/11 · VW-01/03/04 · SCN-01..03 · HLT-01/02 (HC-01..07) · CUS-01..03 · PCK-01/03 · EXP-02/03/04/08/09	5 design partners use it on live work; editable PPTX validated
3. Differentiate → *Pro launch*	10–14	IMP-08/09/10 · SCN-04/05 · HLT-03 · CTL-01..03 · CUS-05 · CMP-01..04 · PCK-02 · EXP-05/06 · AI-01..09	Test plans used weekly by ≥ 3 partners; first paid conversions; AI eval thresholds met
4. Team & Enterprise → *Team GA*	15–24	COL-01..03 · ENT-01..04 · AI-10 · HLT-04 · ENT-05 pilot	First 10 paying teams; one security review passed

15. Pricing & packaging (proposed)

Tier	Price (indicative)	Includes
Free	$0, no account for core use	Local import, 3 views, scenarios, health (Info + Warning), PNG/SVG/PDF export, single definition
Pro	$19–25 per user / month	Packs, compare, PPTX / draw.io / BPMN / XLSX exports, full health and controls, test-plan generator, branding, AI Assist (fair use), BYO key
Team	$49 per user / month (min. 5)	Shared library, versioning, share links, comments, workspace AI policy, priority support
Enterprise	Custom	SSO/SCIM, audit log, data residency, self-hosted build, tenant connector, DPA, SLA

Partner programme: discounted Pro licences plus white-label exports for Workday implementation partners.

16. Go-to-market

1. **Design partners (weeks 1–9).** Recruit 5–10 HRIS leads and consultants, for example people who engaged with the Nov 2026 LinkedIn launch. Exchange early access for anonymised exports, which become the golden corpus.
2. **Build in public.** Weekly LinkedIn clips, one differentiator each, e.g. "14 test cases for Change Job in 10 seconds".
3. **Trust kit.** One-page security note, a network-capture video showing that nothing is uploaded, and a DPA template.
4. **Community.** Workday Community groups, regional user groups and Workday Rising side events; explore a partner or marketplace listing.
5. **Partner channel.** White-label licences for boutique Workday partners.
6. **Content SEO.** "How to read a Workday Change Job definition" and similar guides, with the tool embedded.

17. Analytics & instrumentation

Event	Properties (no content, no file names)	Used for
`import_completed`	steps_count, stages_count, conditions_count, warnings_count, duration_ms, source_type	Activation, quality
`parse_warning`	warning_code	Parser roadmap
`view_changed`	view	G3
`scenario_applied`	conditions_set_count	Engagement
`health_viewed` / `finding_dismissed`	rule_id, severity	Rule tuning
`export_completed`	format, view, pages	North star
`test_plan_generated`	cases_count	G4
`ai_request`	feature, tokens, latency, accepted	AI value and cost

18. Risks & mitigations

Risk	Likelihood	Impact	Mitigation
Export format varies by tenant, release or locale	High	High	Golden corpus from partners; header-signature detection; mapping UI; telemetry on warning codes
Security objections to any cloud use	High	High	Local-first default; AI opt-in with redaction preview; self-hosting; trust kit
Free competitor improves quickly	Medium	Medium	Compete on test plans, controls, compare, BPMN and collaboration rather than the diagram
Trademark complaint	Low	High	Rename; nominative use; disclaimer; no logo
AI output is wrong or misleading	Medium	Medium	Grounding, citations, labelled drafts, eval sets, never writes to the model
Scope creep for a small team	High	Medium	Strict phase exit criteria; public beta before Phase 3
Condition-rule and security-policy exports unavailable or inconsistent	Medium	Medium	Validate with partners before committing; treat as enrichment, not a dependency

19. Open questions & assumptions

#	Question / assumption	Owner	Due
Q1	How much do the export columns vary across releases and locales? Need ≥ 20 samples	Product	Phase 1, week 2
Q2	Can the condition-rule logic and the BP security policy be exported in a consistent format?	Product + partners	Phase 2

#	Question / assumption	Owner	Due
Q3	Should basic downloads need an account (lead capture vs. friction)? Proposed: no	Product	Phase 2
Q4	Final product name and domain; trademark search	Founder	Phase 0
Q5	AI provider and region strategy (EU / US)	Engineering	Phase 3
Q6	Is a Workday partner or marketplace listing feasible and worthwhile?	Founder	Phase 3
Q7	Is RaaS-based tenant sync feasible and acceptable to security teams?	Engineering + partner	Phase 4
A1	*Assumption:* the *View Definition* Excel export is available to the target users	—	Validate with partners
A2	*Assumption:* Workday-delivered default definitions are suitable non-sensitive test fixtures	—	Validate

Appendix A — Prototype issue register

ID	Area	Issue	Location	Fix (phase)
D1	Semantics	False condition drawn as "revert to initiation"	`FlowchartEditor.tsx:137–157, 501–503`	MOD-04 · Phase 0 hot-fix, Phase 1
D2	Semantics	Order letters ignored; no parallel steps	`FlowchartEditor.tsx:82–206`	MOD-03 · Phase 1
D3	Semantics	Only 4 step types mapped	`FlowchartEditor.tsx:89–96`	MOD-05 · Phase 1
D4	Semantics	Non-action nodes show the type name, not the task or group	`FlowchartEditor.tsx:165`	VIS-01 · Phase 1
D5	Layout	Unknown group placed at y=100, overlapping lane 0	`FlowchartEditor.tsx:86–87`	VIS-05 · Phase 1
D6	Model	No validations, notifications, sub-process or routing	`types.ts`	MOD-01 · Phase 1
E1	Security	API key read in client code	`gemini.ts:4` , `vite.config.ts`	Phase 0
E2	Privacy	Tenant data sent to an LLM without consent	`gemini.ts`	Phase 0 (disable) · AI-01/02 in Phase 3
E3	Build	"Add Row" omits required fields (likely TS error); fields not editable	`DataVerification.tsx:20–35`	Phase 0
E4	Export	PNG captures the viewport only	`FlowchartEditor.tsx:342–371`	EXP-01 · Phase 1
E5	State	Canvas and table are separate sources of truth	`FlowchartEditor.tsx:211–235`	MOD-01, CUS-01
E6	State	One-level undo; state mutated in place	`FlowchartEditor.tsx:219–229, 308, 321`	CUS-04
E7	Robustness	LLM output not validated at runtime	`gemini.ts:86`	MOD-01
E8	UX	PDF upload previewed with `<img>` ; single file only	`UploadSection.tsx:29` , `App.tsx:234`	IMP-04
E9	Persistence	No save/load; refresh loses work	—	IMP-07, NFR-REL
E10	Legal	Product name and hotlinked Workday logo	`App.tsx:156–165`	Phase 0
E11	Hygiene	No git, no tests, unused dependencies, AI Studio leftovers	—	Phase 0
E12	Accessibility	Colour-only meaning; no keyboard editing	`CustomNodes.tsx`	VIS-10, NFR-A11Y

Appendix B — Glossary

Term	Meaning
BP / business process	A Workday workflow (e.g. Hire, Change Job, Create Journal)
BP definition	The configured steps of a BP type; there may be a default definition and organisation-specific ones

Term	Meaning
Order	The letter (a, b, c…) that sequences steps; the same letter means the steps run in parallel
Entry condition	A rule on a step; if false the step is skipped
Security group	The group of users who receive a step (the swimlane)
Subprocess	A BP that runs as a step of another BP (e.g. Propose Compensation inside Hire)
Send back / deny / cancel / rescind / correct	Exception actions on an in-flight or completed BP, governed by the BP security policy
Overlay	User customisations stored separately from the imported model
SoD	Segregation of duties: the same person should not initiate and approve
RaaS	Report-as-a-Service: a Workday custom report exposed as a web service
CAB	Change Advisory Board

End of document.